



Database Management Systems

Database Modification

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Database Management Systems

Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

Special Acknowledgement to all CSE Professors who contributed to this endeavor

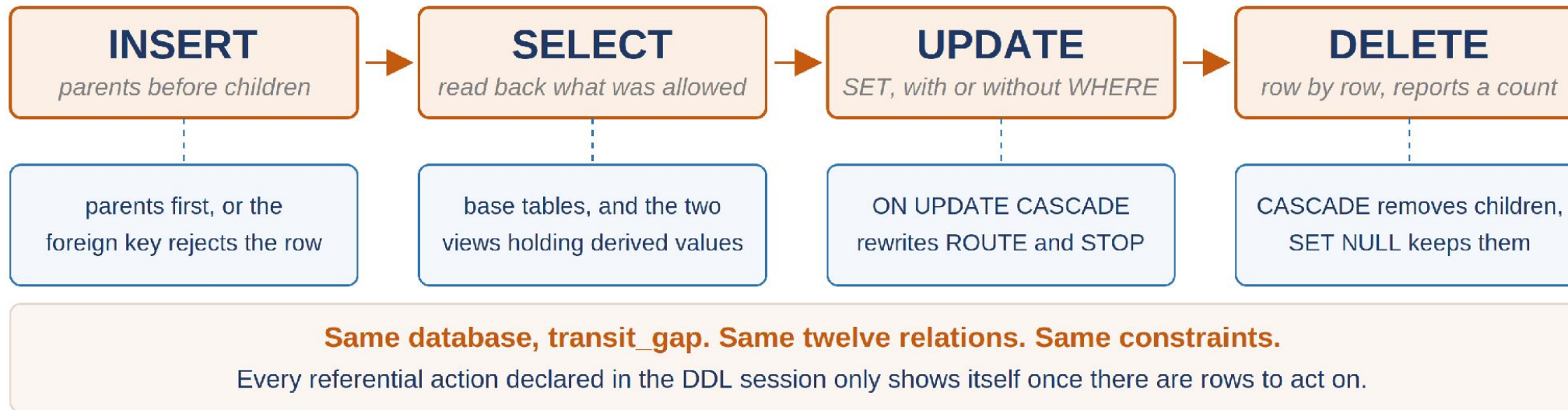
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Database Management Systems

Database modification - the four DML commands



The DDL session built the schema. This one puts data into it.

DML commands change the data held in a schema object: they access, store, modify, update and remove existing records. The order matters - parents before children on the way in, children before parents on the way out - and every referential action declared in the DDL only becomes visible here.

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The twelve relations - keys at a glance



The DDL session created these. This class puts rows into them, and the foreign keys decide the order.

RELATION	PRIMARY KEY	FOREIGN KEYS
AGENCY	Agency_ID	-
WARD	Ward_ID	-
ROUTE	Route_ID	Agency_ID → AGENCY Trunk_Route_ID → ROUTE
STOP	Agency_ID + Stop_Code	Agency_ID → AGENCY Ward_ID → WARD
PLATFORM	Agency_ID + Stop_Code + Platform_Number	Agency_ID + Stop_Code → STOP
SERVES	Route_ID + Agency_ID + Stop_Code	Route_ID → ROUTE Agency_ID + Stop_Code → STOP
PASS_TYPE	Pass_Type_ID	Agency_ID → AGENCY
COMMUTER	Commuter_ID	Ward_ID → WARD
COMMUTER_PHONE	Commuter_ID + Phone_Number	Commuter_ID → COMMUTER
HOLDS	Commuter_ID + Pass_Type_ID	Commuter_ID → COMMUTER Pass_Type_ID → PASS_TYPE
JOURNEY	Journey_ID	Commuter_ID → COMMUTER
FARE_RECEIPT	Receipt_ID	Journey_ID → JOURNEY (UNIQUE)

Two views carry the computed values: COMMUTER_AGE (from Date_of_Birth) and WARD_DENSITY (from Population and Area_SqKm).

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Setup - a clean database



Rebuild the twelve relations first - section 0 of the lab SQL file, from the DDL session

```
prompt (PES1UG23CS024@\\U) [\\d] >\\_
```

```
DROP DATABASE IF EXISTS transit_gap;
```

```
CREATE DATABASE transit_gap;
```

```
USE transit_gap;
```

DROP DATABASE first means the whole script can be re-pasted at any point without hitting 'table already exists'. Run section 0, then start the DML steps.

```
MySQL 8.0 Command Line Cli x + v
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> -- =====
mysql> -- Database Management Systems - Department of CSE
mysql> -- Case study for slides - 03
mysql> -- ER to Relational Schema Mapping and DDL Query Execution
mysql> -- Transit Gap Finder: BMTC + Metro Accessibility Scoring
mysql> --
mysql> -- SRN : PES1UG23CS024
mysql> -- File: PES1UG23CS024_DDL_Lab3.sql
mysql> --
mysql> -- Paste one block at a time and screenshot the result.
mysql> -- Block headings match the slide titles in DDL_Demo.pptx.
mysql> -- Tested on MySQL 8.0 (InnoDB).
mysql> -- =====
mysql> -- DEMO SETUP - session prompt and database
```

```
mysql> -- =====
mysql> -- DEMO SETUP - session prompt and database
mysql> -- =====
mysql> -- NOTE: "prompt" is a mysql client command, not SQL. It works when you
mysql> -- paste it at the mysql> prompt. It will NOT work inside SOURCE.
mysql>
mysql> prompt (PES1UG23CS024@\\U) [\\d] >\\_
PROMPT set to '(PES1UG23CS024@\\U) [\\d] >\\_'
(PES1UG23CS024@root@localhost) [(none)] >
(PES1UG23CS024@root@localhost) [(none)] > CREATE DATABASE IF NOT EXISTS transit_gap;
Query OK, 1 row affected (0.01 sec)

(PES1UG23CS024@root@localhost) [(none)] > USE transit_gap;
Database changed
(PES1UG23CS024@root@localhost) [transit_gap] > SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| movies |
| mysql |
| performance_schema |
| sakila |
| smart_hostel |
| sys |
| transit_gap |
| world |
+-----+
9 rows in set (0.02 sec)
```

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INSERT - form 1, values in column order



No column list: the values must match CREATE TABLE order exactly

```
INSERT INTO AGENCY
VALUES ('BMTCL','Bengaluru Metropolitan Transport Corporation',
       'BUS','Shanthinagar',6000);
```

```
INSERT INTO AGENCY
VALUES ('BMRCL','Bangalore Metro Rail Corporation Ltd',
       'METRO','Swami Vivekananda Road',350);
```

```
SELECT * FROM AGENCY;
```

```
MySQL 8.0 Command Line Cli  x  +  v
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - form 1, values in column order
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO AGENCY
-> VALUES ('BMTCL', 'Bengaluru Metropolitan Transport Corporation',
->         'BUS', 'Shanthinagar', 6000);
Query OK, 1 row affected (0.01 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO AGENCY
-> VALUES ('BMRCL', 'Bangalore Metro Rail Corporation Ltd',
->         'METRO', 'Swami Vivekananda Road', 350);
Query OK, 1 row affected (0.00 sec)

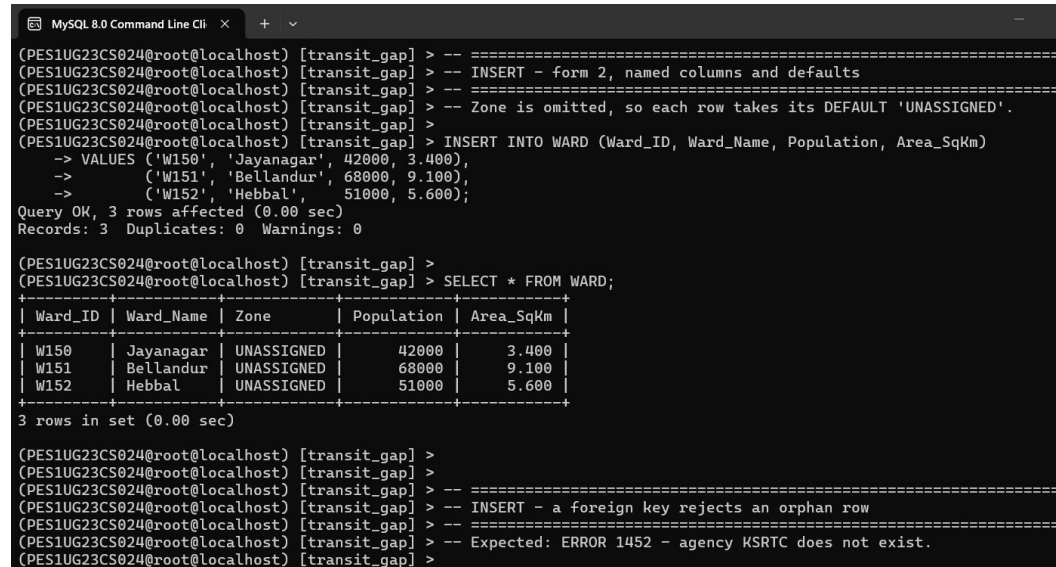
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM AGENCY;
+-----+-----+-----+-----+-----+
| Agency_ID | Agency_Name | Mode | Headquarters_Location | Fleet_Size |
+-----+-----+-----+-----+-----+
| BMRCL     | Bangalore Metro Rail Corporation Ltd | METRO | Swami Vivekananda Road | 350 |
| BMTCL     | Bengaluru Metropolitan Transport Corporation | BUS | Shanthinagar | 6000 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - form 2, named columns and defaults
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
```

Name the columns and the rest take their DEFAULT or NULL

```
INSERT INTO WARD (Ward_ID, Ward_Name, Population, Area_SqKm)
VALUES ('W150', 'Jayanagar', 42000, 3.400),
      ('W151', 'Bellandur', 68000, 9.100),
      ('W152', 'Hebbal', 51000, 5.600);

-- Zone was omitted, so it takes its DEFAULT 'UNASSIGNED'
SELECT * FROM WARD;
```



```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - form 2, named columns and defaults
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- Zone is omitted, so each row takes its DEFAULT 'UNASSIGNED'.
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO WARD (Ward_ID, Ward_Name, Population, Area_SqKm)
-> VALUES ('W150', 'Jayanagar', 42000, 3.400),
->          ('W151', 'Bellandur', 68000, 9.100),
->          ('W152', 'Hebbal', 51000, 5.600);
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD;
+-----+-----+-----+-----+-----+
| Ward_ID | Ward_Name | Zone | Population | Area_SqKm |
+-----+-----+-----+-----+-----+
| W150 | Jayanagar | UNASSIGNED | 42000 | 3.400 |
| W151 | Bellandur | UNASSIGNED | 68000 | 9.100 |
| W152 | Hebbal | UNASSIGNED | 51000 | 5.600 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

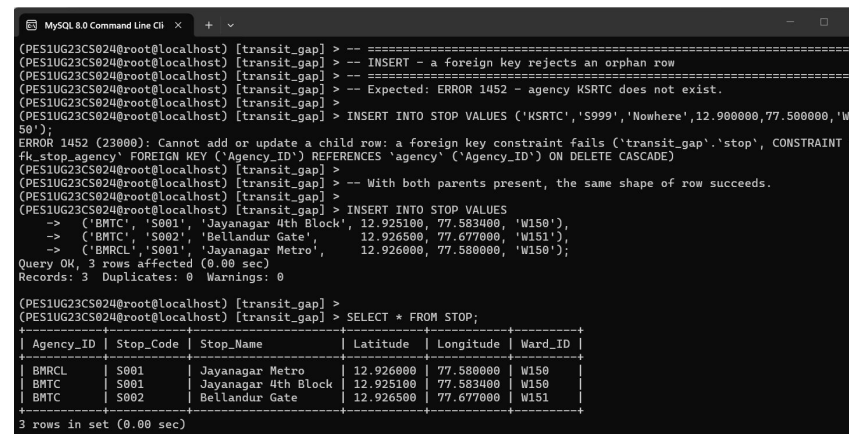
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - a foreign key rejects an orphan row
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 1452 - agency KSRTC does not exist.
(PES1UG23CS024@root@localhost) [transit_gap] >
```

STOP references AGENCY and WARD, so both parents must exist first

```
INSERT INTO STOP VALUES ('KSRTC','S999','Nowhere',12.9,77.5,'W150');  
ERROR 1452 (23000): Cannot add or update a child row:  
a foreign key constraint fails (fk_stop_agency)
```

Insert the parents first, then the child succeeds

```
INSERT INTO STOP VALUES  
('BMTCL','S001','Jayanagar 4th Block',12.925100,77.583400,'W150'),  
('BMTCL','S002','Bellandur Gate',12.926500,77.677000,'W151'),  
('BMRCL','S001','Jayanagar Metro',12.926000,77.580000,'W150');
```



```
MySQL 8.0 Command Line Client  
(PES1UG23CS024@root@localhost) [transit_gap] > --  
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - a foreign key rejects an orphan row  
(PES1UG23CS024@root@localhost) [transit_gap] > --  
(PES1UG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 1452 - agency KSRTC does not exist.  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO STOP VALUES ('KSRTC','S999','Nowhere',12.900000,77.500000,'W150');  
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('transit_gap'. 'stop', CONSTRAINT 'fk_stop_agency' FOREIGN KEY ('Agency_ID') REFERENCES 'agency' ('Agency_ID') ON DELETE CASCADE)  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > -- With both parents present, the same shape of row succeeds.  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO STOP VALUES  
-> ('BMTCL','S001','Jayanagar 4th Block',12.925100,77.583400,'W150'),  
-> ('BMTCL','S002','Bellandur Gate',12.926500,77.677000,'W151'),  
-> ('BMRCL','S001','Jayanagar Metro',12.926000,77.580000,'W150');  
Query OK, 3 rows affected (0.00 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM STOP;  
+-----+-----+-----+-----+-----+-----+  
| Agency_ID | Stop_Code | Stop_Name | Latitude | Longitude | Ward_ID |  
+-----+-----+-----+-----+-----+-----+  
| BMRCL | S001 | Jayanagar Metro | 12.926000 | 77.580000 | W150 |  
| BMTCL | S001 | Jayanagar 4th Block | 12.925100 | 77.583400 | W150 |  
| BMTCL | S002 | Bellandur Gate | 12.926500 | 77.677000 | W151 |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```

PLATFORM hangs off the two-part key of STOP

```
INSERT INTO PLATFORM VALUES
```

```
  ('BMTCL','S001','1','Y'), ('BMTCL','S001','2','N'),  
  ('BMRCL','S001','1','Y'), ('BMRCL','S001','2','Y');
```

Note the same Stop_Code S001 under two different agencies

```
SELECT * FROM PLATFORM;
```

```
SELECT Agency_ID, Stop_Code, COUNT(*) AS platforms  
FROM PLATFORM GROUP BY Agency_ID, Stop_Code;
```

```
MySQL 8.0 Command Line CL  x  +  v  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - the weak entity's own children  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > -- Stop_Code S001 exists under two agencies; the three-part key keeps  
(PES1UG23CS024@root@localhost) [transit_gap] > -- their platforms apart.  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO PLATFORM VALUES  
-> ('BMTCL','S001','1','Y'),  
-> ('BMTCL','S001','2','N'),  
-> ('BMRCL','S001','1','Y'),  
-> ('BMRCL','S001','2','Y');  
Query OK, 4 rows affected (0.00 sec)  
Records: 4  Duplicates: 0  Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM PLATFORM;  
+-----+-----+-----+-----+  
| Agency_ID | Stop_Code | Platform_Number | Wheelchair_Access |  
+-----+-----+-----+-----+  
| BMRCL     | S001      | 1               | Y                  |  
| BMRCL     | S001      | 2               | Y                  |  
| BMTCL     | S001      | 1               | Y                  |  
| BMTCL     | S001      | 2               | N                  |  
+-----+-----+-----+-----+  
4 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >
```

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INSERT - routes, including the self-referencing column

INSERT INTO ROUTE VALUES

```
('R500','500','Silk Board - Hebbal','TRUNK',24.50,'BMTCL',NULL),
('R501','501A','Jayanagar Feeder','FEEDER',6.20,'BMTCL','R500'),
('R502','502B','Bellandur Feeder','FEEDER',7.80,'BMTCL','R500'),
('RM01','PURPLE','Purple Line','METRO',43.00,'BMRCL',NULL);
```

Two feeders point at the trunk route; the trunk points at nothing

SELECT Route_ID, Route_Number, Trunk_Route_ID FROM ROUTE;

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Agency_ID, Stop_Code, COUNT(*) AS platforms
  -> FROM PLATFORM
  -> GROUP BY Agency_ID, Stop_Code;
+-----+-----+-----+
| Agency_ID | Stop_Code | platforms |
+-----+-----+-----+
| BMRCL     | S001     | 2         |
| BMTCL     | S001     | 2         |
+-----+-----+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - routes, including the recursive column
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - the M:N relation carries its own attributes
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT INTO ROUTE VALUES
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO ROUTE VALUES
  -> ('R500','500','Silk Board - Hebbal','TRUNK',24.50,'BMTCL',NULL),
  -> ('R501','501A','Jayanagar Feeder','FEEDER',6.20,'BMTCL','R500'),
  -> ('R502','502B','Bellandur Feeder','FEEDER',7.80,'BMTCL','R500'),
  -> ('RM01','PURPLE','Purple Line','METRO',43.00,'BMRCL',NULL);
Query OK, 4 rows affected (0.00 sec)
Records: 4 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_ID, Route_Number, Trunk_Route_ID FROM ROUTE;
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_ID, Route_Number, Trunk_Route_ID FROM ROUTE;
+-----+-----+-----+
| Route_ID | Route_Number | Trunk_Route_ID |
+-----+-----+-----+
| R500     | 500         | NULL            |
| R501     | 501A        | R500            |
| R502     | 502B        | R500            |
| RM01     | PURPLE      | NULL            |
+-----+-----+-----+
4 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - the M:N relation carries its own attributes
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT INTO SERVES VALUES
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO SERVES VALUES
  -> ('R500','BMTCL','S001',1,0,0.00),
  -> ('R500','BMTCL','S002',2,18,9.40),
  -> ('R501','BMTCL','S001',1,0,0.00);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM SERVES;
+-----+-----+-----+-----+-----+-----+
| Route_ID | Agency_ID | Stop_Code | Stop_Sequence | Scheduled_Offset_Minutes | Distance_From-Origin_KM |
+-----+-----+-----+-----+-----+-----+
| R500     | BMTCL     | S001     | 1             | 0                       | 0.00                    |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

SERVES holds Stop_Sequence, the offset and the distance

```
INSERT INTO SERVES VALUES
```

```
('R500','BMT','S001',1, 0, 0.00),  
( 'R500','BMT','S002',2,18, 9.40),  
( 'R501','BMT','S001',1, 0, 0.00);
```

uq_serves_sequence stops the same position being used twice on a route

```
INSERT INTO SERVES VALUES ('R500','BMRCL','S001',1,0,0.00);  
ERROR 1062 (23000): Duplicate entry 'R500-1' for key  
'serves.uq_serves_sequence'
```

```
MySQL 8.0 Command Line Client
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM SERVES;
+-----+-----+-----+-----+-----+-----+
| Route_ID | Agency_ID | Stop_Code | Stop_Sequence | Scheduled_Offset_Minutes | Distance_From_Origin_KM |
+-----+-----+-----+-----+-----+-----+
| R500     | BMT       | S001      | 1             | 0                         | 0.00                    |
| R500     | BMT       | S002      | 2             | 18                        | 9.40                    |
| R501     | BMT       | S001      | 1             | 0                         | 0.00                    |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 1062 - uq_serves_sequence already has position 1
(PES1UG23CS024@root@localhost) [transit_gap] > -- on route R500.
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO SERVES VALUES ('R500','BMRCL','S001',1,0,0.00);
(PES1UG23CS024@root@localhost) [transit_gap] >
ERROR 1062 (23000): Duplicate entry 'R500-1' for key 'serves.uq_serves_sequence'
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - commuters, phones and passes
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO COMMUTER VALUES
-> ('C001','Asha', NULL, 'Rao', 'F', '2001-04-12', 'asha@example.com', 'W150'),
-> ('C002','Vikram', 'N', 'Iyer', 'M', '1998-09-30', 'vikram@example.com', 'W151');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO COMMUTER_PHONE VALUES
```

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INSERT - commuters, phones and passes



```
INSERT INTO COMMUTER VALUES
```

```
  ('C001','Asha',NULL,'Rao','F','2001-04-12','asha@example.com','W150'),  
  ('C002','Vikram','N','Iyer','M','1998-09-30','vikram@example.com','W151');
```

```
INSERT INTO COMMUTER_PHONE VALUES
```

```
  ('C001','9845010001'), ('C001','8025550002'), ('C002','9845010003');
```

```
INSERT INTO PASS_TYPE VALUES
```

```
  ('P01','Daily Bus Pass', 1, 70.00,'BMTCL'),  
  ('P02','Monthly Bus Pass',30,1200.00,'BMTCL'),  
  ('P03','Metro Smart Card',90, 500.00,'BMRCL');
```

```
INSERT INTO HOLDS VALUES ('C001','P02'), ('C001','P03'), ('C002','P01');
```

```
MySQL 8.0 Command Line Cli x + -  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO COMMUTER_PHONE VALUES  
-> ('C001','9845010001'),  
-> ('C001','8025550002'),  
-> ('C002','9845010003');  
Query OK, 3 rows affected (0.00 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO PASS_TYPE VALUES  
-> ('P01','Daily Bus Pass', 1, 70.00, 'BMTCL'),  
-> ('P02','Monthly Bus Pass', 30, 1200.00, 'BMTCL'),  
-> ('P03','Metro Smart Card', 90, 500.00, 'BMRCL');  
Query OK, 3 rows affected (0.00 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO HOLDS VALUES ('C001','P02'), ('C001','P03'), ('C002','P01');  
Query OK, 3 rows affected (0.00 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM COMMUTER;  
+-----+  
| Commuter_ID | First_Name | Middle_Name | Last_Name | Gender | Date_of_Birth | Email_Address | Ward_ID |  
+-----+  
| C001        | Asha       | NULL        | Rao       | F      | 2001-04-12    | asha@example.com | W150    |  
| C002        | Vikram     | N           | Iyer      | M      | 1998-09-30    | vikram@example.com | W151    |  
+-----+  
2 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli x + -  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM COMMUTER_PHONE;  
+-----+  
| Commuter_ID | Phone_Number |  
+-----+  
| C001        | 8025550002   |  
| C001        | 9845010001   |  
| C002        | 9845010003   |  
+-----+  
3 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM PASS_TYPE;  
+-----+  
| Pass_Type_ID | Pass_Name      | Validity_Days | Price | Agency_ID |  
+-----+  
| P01          | Daily Bus Pass | 1             | 70.00 | BMTCL     |  
| P02          | Monthly Bus Pass | 30            | 1200.00 | BMTCL    |  
| P03          | Metro Smart Card | 90            | 500.00 | BMRCL     |  
+-----+  
3 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM HOLDS;  
+-----+  
| Commuter_ID | Pass_Type_ID |  
+-----+  
| C002        | P01          |  
| C001        | P02          |  
| C001        | P03          |  
+-----+  
3 rows in set (0.00 sec)
```

```
INSERT INTO JOURNEY VALUES
```

```
  ('J001','2026-08-20','08:05:00','08:47:00',1,'C001'),  
  ('J002','2026-08-20','18:10:00','18:55:00',0,'C002');
```

```
INSERT INTO FARE_RECEIPT VALUES ('RC01','2026-08-20',45.00,'J001');
```

A second receipt for the same journey breaks uq_receipt_journey

```
INSERT INTO FARE_RECEIPT VALUES ('RC02','2026-08-20',45.00,'J001');
```

```
ERROR 1062 (23000): Duplicate entry 'J001' for key  
'fare_receipt.uq_receipt_journey'
```

```
MySQL 8.0 Command Line Cli  x  +  v  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT - the 1:1 pair, and what UNIQUE forbids  
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO JOURNEY VALUES  
-> ('J001','2026-08-20','08:05:00','08:47:00',1,'C001');  
-> ('J002','2026-08-20','18:10:00','18:55:00',0,'C002');  
Query OK, 2 rows affected (0.00 sec)  
Records: 2 Duplicates: 0 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO FARE_RECEIPT VALUES ('RC01','2026-08-20',45.00,'J001');  
Query OK, 1 row affected (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 1062 - uq_receipt_journey. This is what makes  
(PES1UG23CS024@root@localhost) [transit_gap] > -- GENERATES a 1:1 relationship rather than 1:N.  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO FARE_RECEIPT VALUES ('RC02','2026-08-20',45.00,'J001');  
ERROR 1062 (23000): Duplicate entry 'J001' for key 'fare_receipt.uq_receipt_journey'  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM FARE_RECEIPT;  
  
+-----+-----+-----+-----+  
| Receipt_ID | Receipt_Date | Fare_Amount | Journey_ID |  
+-----+-----+-----+-----+  
| RC01       | 2026-08-20   | 45.00       | J001       |  
+-----+-----+-----+-----+  
1 row in set (0.00 sec)
```

Database Management Systems

INSERT - the CHECK and NOT NULL constraints firing



```
INSERT INTO AGENCY VALUES ('X','X Transit','TRAM','Nowhere',10);
ERROR 3819 (HY000): Check constraint 'chk_agency_mode' is violated.
```

```
INSERT INTO WARD (Ward_ID, Ward_Name, Area_SqKm)
VALUES ('W153','Nowhere Ward',0);
ERROR 3819 (HY000): Check constraint 'chk_ward_area' is violated.
```

```
INSERT INTO WARD (Ward_ID, Population, Area_SqKm)
VALUES ('W154',1000,2.0);
ERROR 1364 (HY000): Field 'Ward_Name' doesn't have a default value
```

```
MySQL 8.0 Command Line Cli  x  +  -  x
(PESIUG23CS024@root@localhost) [transit_gap] > =====
(PESIUG23CS024@root@localhost) [transit_gap] > -- INSERT - the CHECK and NOT NULL constraints firing
(PESIUG23CS024@root@localhost) [transit_gap] > =====
(PESIUG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 3819 - chk_agency_mode allows only BUS or METRO.
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > INSERT INTO AGENCY VALUES ('X','X Transit','TRAM','Nowhere',10);
ERROR 3819 (HY000): Check constraint 'chk_agency_mode' is violated.
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 3819 - chk_ward_area requires Area_SqKm > 0.
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > INSERT INTO WARD (Ward_ID, Ward_Name, Area_SqKm) VALUES ('W153','Nowhere
Ward',0);
ERROR 3819 (HY000): Check constraint 'chk_ward_area' is violated.
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 1364 - Ward_Name is NOT NULL with no default.
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > INSERT INTO WARD (Ward_ID, Population, Area_SqKm) VALUES ('W154',1000,2.0
);
ERROR 1364 (HY000): Field 'Ward_Name' doesn't have a default value
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > -- =====
(PESIUG23CS024@root@localhost) [transit_gap] > -- INSERT ... SELECT - load a table from a query
(PESIUG23CS024@root@localhost) [transit_gap] > =====
(PESIUG23CS024@root@localhost) [transit_gap] >
(PESIUG23CS024@root@localhost) [transit_gap] > CREATE TABLE ROUTE_STOP_INFO (
-> Route_Number VARCHAR(15)
-> Stop_Name VARCHAR(120),
-> Stop_Sequence INT
-> );
```

Database Management Systems

INSERT ... SELECT - load a table from a query

Create the target first, then fill it from a query

```
CREATE TABLE ROUTE_STOP_INFO (  
    Route_Number VARCHAR(15),  
    Stop_Name     VARCHAR(120),  
    Stop_Sequence INT);  
  
INSERT INTO ROUTE_STOP_INFO (Route_Number, Stop_Name, Stop_Sequence)  
SELECT R.Route_Number, S.Stop_Name, V.Stop_Sequence  
FROM   ROUTE R, SERVES V, STOP S  
WHERE  R.Route_ID = V.Route_ID  
       AND V.Agency_ID = S.Agency_ID AND V.Stop_Code = S.Stop_Code;  
  
SELECT * FROM ROUTE_STOP_INFO;
```

```
MySQL 8.0 Command Line Cli x + -  
(PES1UG23CS024@root@localhost) [transit_gap] > --  
(PES1UG23CS024@root@localhost) [transit_gap] > -- INSERT ... SELECT - load a table from a query  
(PES1UG23CS024@root@localhost) [transit_gap] > --  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE ROUTE_STOP_INFO (  
(PES1UG23CS024@root@localhost) [transit_gap] >     Route_Number VARCHAR(15),  
->     Stop_Name VARCHAR(120),  
->     Stop_Sequence INT  
-> );  
Query OK, 0 rows affected (0.01 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO ROUTE_STOP_INFO (Route_Number, Stop_Name, Stop_Sequence)  
-> SELECT R.Route_Number, S.Stop_Name, V.Stop_Sequence  
-> FROM   ROUTE R, SERVES V, STOP S  
-> WHERE  R.Route_ID = V.Route_ID  
->       AND V.Agency_ID = S.Agency_ID  
->       AND V.Stop_Code = S.Stop_Code;  
Query OK, 3 rows affected (0.00 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM ROUTE_STOP_INFO;  
+-----+-----+-----+  
| Route_Number | Stop_Name | Stop_Sequence |  
+-----+-----+-----+  
| 500          | Jayanagar 4th Block | 1 |  
| 500          | Bellandur Gate      | 2 |  
| 501A         | Jayanagar 4th Block | 1 |  
+-----+-----+-----+
```

```
MySQL 8.0 Command Line Cli x + -  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > -- The backup form needs an identically shaped table.  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > CREATE TABLE ROUTE_STOP_INFO_BACKUP LIKE ROUTE_STOP_INFO;  
Query OK, 0 rows affected (0.01 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] > INSERT INTO ROUTE_STOP_INFO_BACKUP (SELECT * FROM ROUTE_STOP_INFO);  
Query OK, 3 rows affected (0.00 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM ROUTE_STOP_INFO_BACKUP;  
+-----+-----+-----+  
| Route_Number | Stop_Name | Stop_Sequence |  
+-----+-----+-----+  
| 500          | Jayanagar 4th Block | 1 |  
| 500          | Bellandur Gate      | 2 |  
| 501A         | Jayanagar 4th Block | 1 |  
+-----+-----+-----+  
3 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] >
```

Database Management Systems

SELECT - reading the data back



```
SELECT Agency_Name, Mode, Fleet_Size FROM AGENCY;
```

```
SELECT Stop_Name, Ward_ID FROM STOP WHERE Agency_ID = 'BMTC';
```

```
SELECT Route_Number, Total_Distance_KM FROM ROUTE  
WHERE Total_Distance_KM > 10;
```

```
-- the computed values, read from the views
```

```
SELECT * FROM COMMUTER_AGE;
```

```
SELECT * FROM WARD_DENSITY;
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- SELECT - reading the data back
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Agency_Name, Mode, Fleet_Size FROM AGENCY;
(PES1UG23CS024@root@localhost) [transit_gap] >
+-----+-----+-----+
| Agency_Name | Mode | Fleet_Size |
+-----+-----+-----+
| Bangalore Metro Rail Corporation Ltd | METRO | 350 |
| Bengaluru Metropolitan Transport Corporation | BUS | 6000 |
+-----+-----+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Stop_Name, Ward_ID FROM STOP WHERE Agency_ID = 'BMTC';
+-----+-----+
| Stop_Name | Ward_ID |
+-----+-----+
| Jayanagar 4th Block | W150 |
| Bellandur Gate | W151 |
+-----+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_Number, Total_Distance_KM FROM ROUTE WHERE Total_Distance_KM > 10;
+-----+-----+
| Route_Number | Total_Distance_KM |
+-----+-----+
| 500 | 24.50 |
| PURPLE | 43.00 |
+-----+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- The derived attributes, recomputed by the views every time.
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM COMMUTER_AGE;
+-----+-----+
| Commuter_ID | Age |
+-----+-----+
| C001 | 25 |
| C002 | 27 |
+-----+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD_DENSITY;
+-----+-----+
| Ward_ID | Population_Density |
+-----+-----+
| W150 | 12352.94 |
| W151 | 7472.53 |
| W152 | 9107.14 |
+-----+-----+
```

Database Management Systems

UPDATE - with a WHERE clause



WHERE selects the rows; **SET** names the columns and their new values

```
UPDATE WARD SET Zone = 'South' WHERE Ward_ID = 'W150';
```

```
UPDATE WARD SET Zone = 'East' WHERE Ward_ID = 'W151';
```

```
SELECT * FROM WARD;
```

Several columns in one statement

```
UPDATE ROUTE
```

```
SET Route_Name = 'Silk Board - Hebbal Express',
```

```
Total_Distance_KM = 25.10
```

```
WHERE Route_ID = 'R500';
```

```
MySQL 8.0 Command Line CL
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD_DENSITY;
+-----+-----+
| Ward_ID | Population_Density |
+-----+-----+
| W150    | 12252.96           |
| W151    | 7472.53             |
| W152    | 9187.14             |
+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > --
(PES1UG23CS024@root@localhost) [transit_gap] > -- UPDATE - with a WHERE clause
(PES1UG23CS024@root@localhost) [transit_gap] > --
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD;
+-----+-----+-----+-----+-----+
| Ward_ID | Ward_Name | Zone | Population | Area_SqKm |
+-----+-----+-----+-----+-----+
| W150    | Jayanagar | UNASSIGNED | 42000 | 3.400 |
| W151    | Bellandur | UNASSIGNED | 68000 | 9.100 |
| W152    | Hebbal    | UNASSIGNED | 51000 | 5.600 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE WARD SET Zone = 'South' WHERE Ward_ID = 'W150';
Query OK, 1 row affected (0.00 sec)
```

```
MySQL 8.0 Command Line CL
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE WARD SET Zone = 'South' WHERE Ward_ID = 'W150';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE WARD SET Zone = 'East' WHERE Ward_ID = 'W151';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD;
+-----+-----+-----+-----+-----+
| Ward_ID | Ward_Name | Zone | Population | Area_SqKm |
+-----+-----+-----+-----+-----+
| W150    | Jayanagar | South | 42000 | 3.400 |
| W151    | Bellandur | East  | 68000 | 9.100 |
| W152    | Hebbal    | UNASSIGNED | 51000 | 5.600 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE ROUTE
-> SET Route_Name = 'Silk Board - Hebbal Express',
-> Total_Distance_KM = 25.10
-> WHERE Route_ID = 'R500';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

Database Management Systems

UPDATE - no WHERE means every row



Leaving out WHERE updates the whole table - rows matched tells you so

```
UPDATE WARD SET Zone = 'UNASSIGNED';
```

```
Query OK, 3 rows affected    Rows matched: 3    Changed: 3
```

An expression can be used on the right of SET

```
UPDATE PASS_TYPE SET Price = Price * 1.10 WHERE Agency_ID = 'BMTC';
```

```
SELECT Pass_Name, Price FROM PASS_TYPE;
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_ID, Route_Name, Total_Distance_KM FROM ROUTE;
+-----+-----+-----+
| Route_ID | Route_Name | Total_Distance_KM |
+-----+-----+-----+
| R500 | Silk Board - Hebbal Express | 25.10 |
| R501 | Jayanagar Feeder | 6.20 |
| R502 | Bellandur Feeder | 7.80 |
| RM01 | Purple Line | 43.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- UPDATE - no WHERE means every row
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- Read 'Rows matched' in the reply, not just 'rows affected'.
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE WARD SET Zone = 'UNASSIGNED';
Query OK, 3 rows affected (0.00 sec)
Rows matched: 3 Changed: 3 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD;
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD;
+-----+-----+-----+-----+-----+
| Ward_ID | Ward_Name | Zone | Population | Area_SqKm |
+-----+-----+-----+-----+-----+
| W150 | Jayanagar | UNASSIGNED | 42000 | 3.400 |
| W151 | Bellandur | UNASSIGNED | 68000 | 9.100 |
| W152 | Hebbal | UNASSIGNED | 51000 | 5.600 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE PASS_TYPE SET Price = Price * 1.10 WHERE Agency_ID = 'BMTC';
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2 Changed: 2 Warnings: 0

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Pass_Name, Price FROM PASS_TYPE;
+-----+-----+
| Pass_Name | Price |
+-----+-----+
| Daily Bus Pass | 77.00 |
| Monthly Bus Pass | 1320.00 |
| Metro Smart Card | 500.00 |
+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- UPDATE - MySQL and the self-referencing subquery
```

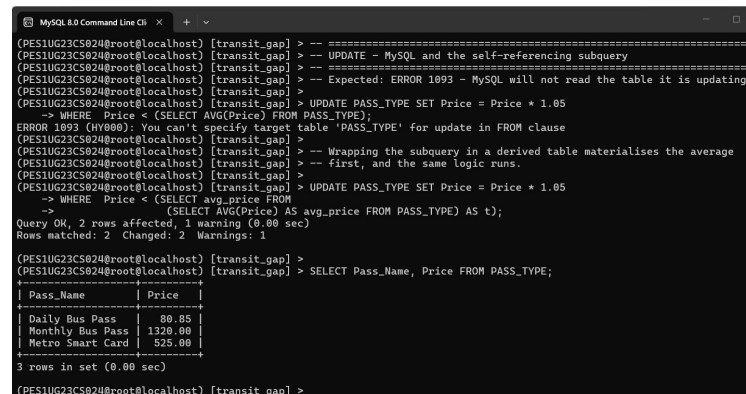
This form is rejected by MySQL

```
UPDATE PASS_TYPE SET Price = Price * 1.05
WHERE Price < (SELECT AVG(Price) FROM PASS_TYPE);
ERROR 1093 (HY000): You can't specify target table 'PASS_TYPE'
for update in FROM clause
```

Wrap the subquery in a derived table and it runs

```
UPDATE PASS_TYPE SET Price = Price * 1.05
WHERE Price < (SELECT avg_price FROM
               (SELECT AVG(Price) AS avg_price FROM PASS_TYPE) AS t);

SELECT Pass_Name, Price FROM PASS_TYPE;
```



```
MySQL 8.0 Command Line C#
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- UPDATE - MySQL and the self-referencing subquery
(PES1UG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 1093 - MySQL will not read the table it is updating.
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE PASS_TYPE SET Price = Price * 1.05
(PES1UG23CS024@root@localhost) [transit_gap] > --> WHERE Price < (SELECT AVG(Price) FROM PASS_TYPE);
ERROR 1093 (HY000): You can't specify target table 'PASS_TYPE' for update in FROM clause
(PES1UG23CS024@root@localhost) [transit_gap] > -- Wrapping the subquery in a derived table materialises the average
(PES1UG23CS024@root@localhost) [transit_gap] > -- first, and the same logic runs.
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE PASS_TYPE SET Price = Price * 1.05
(PES1UG23CS024@root@localhost) [transit_gap] > --> WHERE Price < (SELECT avg_price FROM
(PES1UG23CS024@root@localhost) [transit_gap] > (SELECT AVG(Price) AS avg_price FROM PASS_TYPE) AS t);
Query OK, 2 rows affected, 1 warning (0.08 sec)
Rows matched: 2 Changed: 2 Warnings: 1

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Pass_Name, Price FROM PASS_TYPE;
+-----+-----+
| Pass_Name | Price |
+-----+-----+
| Daily Bus Pass | 80.85 |
| Monthly Bus Pass | 1220.00 |
| Metro Smart Card | 525.00 |
+-----+-----+
3 rows in set (0.08 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
```

fk_route_agency and fk_stop_agency were declared ON UPDATE CASCADE

```
SELECT Route_ID, Agency_ID FROM ROUTE;  
SELECT Agency_ID, Stop_Code FROM STOP;
```

```
UPDATE AGENCY SET Agency_ID = 'BMTCL-N' WHERE Agency_ID = 'BMTCL';
```

The children were rewritten without being touched

```
SELECT Route_ID, Agency_ID FROM ROUTE;  
SELECT Agency_ID, Stop_Code FROM STOP;
```

```
UPDATE AGENCY SET Agency_ID = 'BMTCL' WHERE Agency_ID = 'BMTCL-N';
```

```
MySQL 8.0 Command Line CL - x + -  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_ID, Agency_ID FROM ROUTE;  
+-----+-----+  
| Route_ID | Agency_ID |  
+-----+-----+  
| RM01    | BMRCL    |  
| RS01    | BMTCL    |  
| RS02    | BMTCL    |  
+-----+-----+  
3 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Agency_ID, Stop_Code FROM STOP;  
+-----+-----+  
| Agency_ID | Stop_Code |  
+-----+-----+  
| BMRCL     | S001      |  
| BMTCL     | S001      |  
| BMTCL     | S002      |  
+-----+-----+  
3 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE AGENCY SET Agency_ID = 'BMTCL-N' WHERE Agency_ID = 'BMTCL';  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_ID, Agency_ID FROM ROUTE;  
+-----+-----+  
| Route_ID | Agency_ID |  
+-----+-----+  
| RM01    | BMRCL    |  
| RS01    | BMTCL-N  |  
| RS02    | BMTCL-N  |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line CL - x + -  
+-----+-----+  
| Agency_ID | Stop_Code |  
+-----+-----+  
| BMRCL     | S001      |  
| BMTCL     | S001      |  
| BMTCL     | S002      |  
+-----+-----+  
3 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE AGENCY SET Agency_ID = 'BMTCL-N' WHERE Agency_ID = 'BMTCL';  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_ID, Agency_ID FROM ROUTE;  
+-----+-----+  
| Route_ID | Agency_ID |  
+-----+-----+  
| RM01    | BMRCL    |  
| RS01    | BMTCL-N  |  
| RS02    | BMTCL-N  |  
+-----+-----+  
3 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Agency_ID, Stop_Code FROM STOP;  
+-----+-----+  
| Agency_ID | Stop_Code |  
+-----+-----+  
| BMRCL     | S001      |  
| BMTCL-N   | S001      |  
| BMTCL-N   | S002      |  
+-----+-----+  
3 rows in set (0.00 sec)  
  
(PES1UG23CS024@root@localhost) [transit_gap] >  
(PES1UG23CS024@root@localhost) [transit_gap] > UPDATE AGENCY SET Agency_ID = 'BMTCL' WHERE Agency_ID = 'BMTCL-N';  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
(PES1UG23CS024@root@localhost) [transit_gap] >
```

Database Management Systems

DELETE - with a WHERE clause



```
SELECT * FROM COMMUTER_PHONE;
```

```
DELETE FROM COMMUTER_PHONE
```

```
WHERE Commuter_ID = 'C001' AND Phone_Number = '8025550002';
```

```
SELECT * FROM COMMUTER_PHONE;
```

A WHERE that matches nothing is not an error - it deletes nothing

```
DELETE FROM COMMUTER_PHONE WHERE Commuter_ID = 'C999';
```

Query OK, 0 rows affected

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- DELETE - with a WHERE clause
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM COMMUTER_PHONE;

+-----+
| Commuter_ID | Phone_Number |
+-----+
| C001        | 8025550002   |
| C001        | 9845010001   |
| C002        | 9845010003   |
+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > DELETE FROM COMMUTER_PHONE
  -> WHERE Commuter_ID = 'C001' AND Phone_Number = '8025550002';
Query OK, 1 row affected (0.01 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM COMMUTER_PHONE;

+-----+
| Commuter_ID | Phone_Number |
+-----+
| C001        | 9845010001   |
| C002        | 9845010003   |
+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > -- Matching nothing is not an error.
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > DELETE FROM COMMUTER_PHONE WHERE Commuter_ID = 'C999';
Query OK, 0 rows affected (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- DELETE - ON DELETE CASCADE in effect
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT COUNT(*) AS stops FROM STOP;

+-----+
| stops |
+-----+
| 3     |
+-----+
1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT COUNT(*) AS platforms FROM PLATFORM;

+-----+
| platforms |
+-----+
| 4         |
+-----+
1 row in set (0.01 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT COUNT(*) AS routes FROM ROUTE;

+-----+
| routes |
+-----+
```

One DELETE, four tables changed

```
SELECT COUNT(*) FROM STOP;          SELECT COUNT(*) FROM PLATFORM;
SELECT COUNT(*) FROM ROUTE;          SELECT COUNT(*) FROM SERVES;
```

```
DELETE FROM AGENCY WHERE Agency_ID = 'BMRCL';
```

BMRCL's stops went, and their platforms went with them

```
SELECT * FROM STOP;
SELECT * FROM PLATFORM;
SELECT * FROM ROUTE;
```

```
MySQL 8.0 Command Line Cl. X
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT COUNT(*) FROM STOP;  SELECT COUNT(*) FROM PLATFORM;
+-----+
| COUNT(*) |
+-----+
| 3 |
+-----+
1 row in set (0.00 sec)

+-----+
| COUNT(*) |
+-----+
| 4 |
+-----+
1 row in set (0.02 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT COUNT(*) FROM ROUTE;  SELECT COUNT(*) FROM SERVES;
+-----+
| COUNT(*) |
+-----+
| 3 |
+-----+
1 row in set (0.00 sec)

+-----+
| COUNT(*) |
+-----+
| 1 |
+-----+
1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > DELETE FROM AGENCY WHERE Agency_ID = 'BMRCL';
Query OK, 1 row affected (0.00 sec)
```

```
MySQL 8.0 Command Line Cl. X
+-----+
| COUNT(*) |
+-----+
| 1 |
+-----+
1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > DELETE FROM AGENCY WHERE Agency_ID = 'BMRCL';
Query OK, 1 row affected (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM STOP;
(PES1UG23CS024@root@localhost) [transit_gap] >
+-----+
| Agency_ID | Stop_Code | Stop_Name | Latitude | Longitude | Ward_ID |
+-----+
| BMTCL | S001 | Jayanagar 4th Block | 12.925100 | 77.583400 | W150 |
| BMTCL | S002 | Bellandur Gate | 12.926500 | 77.677000 | W151 |
+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM PLATFORM;
+-----+
| Agency_ID | Stop_Code | Platform_Number | Wheelchair_Access |
+-----+
| BMTCL | S001 | 1 | Y |
| BMTCL | S001 | 2 | N |
+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM ROUTE;
+-----+
| Route_ID | Route_Number | Route_Name | Route_Type | Total_Distance_KM | Agency_ID | Trunk_Route_ID |
+-----+
| R501 | S01A | Jayanagar Feeder | FEEDER | 6.20 | BMTCL | NULL |
| R502 | S02B | Bellandur Feeder | FEEDER | 7.80 | BMTCL | NULL |
+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
```

fk_route_trunk was declared ON DELETE SET NULL, not CASCADE

```
SELECT Route_ID, Trunk_Route_ID FROM ROUTE;
```

```
DELETE FROM ROUTE WHERE Route_ID = 'R500';
```

The feeder routes survive; their pointer is blanked

```
SELECT Route_ID, Trunk_Route_ID FROM ROUTE;
```

Had this been ON DELETE CASCADE, withdrawing one trunk route would have deleted R501 and R502 as well - and then their feeders.

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- DELETE - ON DELETE SET NULL in effect
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_ID, Trunk_Route_ID FROM ROUTE;
+-----+-----+
| Route_ID | Trunk_Route_ID |
+-----+-----+
| R500     | NULL           |
| RM01     | NULL           |
| R501     | R500           |
| R502     | R500           |
+-----+-----+
4 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > DELETE FROM ROUTE WHERE Route_ID = 'R500';
Query OK, 1 row affected (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- R501 and R502 survive as standalone routes with a blanked pointer.
(PES1UG23CS024@root@localhost) [transit_gap] > -- Had fk_route_trunk been declared CASCADE, both would have been
(PES1UG23CS024@root@localhost) [transit_gap] > -- deleted, and then anything feeding them.
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > -- R501 and R502 survive as standalone routes with a blanked pointer.
(PES1UG23CS024@root@localhost) [transit_gap] > -- Had fk_route_trunk been declared CASCADE, both would have been
(PES1UG23CS024@root@localhost) [transit_gap] > -- deleted, and then anything feeding them.
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT Route_ID, Trunk_Route_ID FROM ROUTE;
+-----+-----+
| Route_ID | Trunk_Route_ID |
+-----+-----+
| R501     | NULL           |
| R502     | NULL           |
| RM01     | NULL           |
+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- DELETE without WHERE, and how it differs from TRUNCATE
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT COUNT(*) FROM ROUTE_STOP_INFO;
+-----+
| COUNT(*) |
+-----+
| 3         |
+-----+
1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > DELETE FROM ROUTE_STOP_INFO;
-- row by row, reports the count
```

Database Management Systems

DELETE without WHERE, and how it differs from TRUNCATE



```
SELECT COUNT(*) FROM ROUTE_STOP_INFO;
```

```
DELETE FROM ROUTE_STOP_INFO;          -- row by row, reports the count
```

```
SELECT COUNT(*) FROM ROUTE_STOP_INFO;
```

```
TRUNCATE TABLE ROUTE_STOP_INFO;      -- no row count, resets AUTO_INCREMENT
```

DELETE is DML and reports rows affected. TRUNCATE is DDL, cannot carry a WHERE clause, and is refused on a table a foreign key still references.

```
TRUNCATE TABLE AGENCY;
```

ERROR 1701 (42000): Cannot truncate a table referenced in a foreign key constraint

```
MySQL 8.0 Command Line CL  X  +  -  x
(PES1UG23CS024@root@localhost) [transit_gap] > DELETE FROM ROUTE_STOP_INFO;          -- row by row, reports the count
Query OK, 3 rows affected (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT COUNT(*) FROM ROUTE_STOP_INFO; -- 0 rows, table still exists
+-----+
| COUNT(*) |
+-----+
|         0 |
+-----+
1 row in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > TRUNCATE TABLE ROUTE_STOP_INFO_BACKUP;
Query OK, 0 rows affected (0.03 sec)

(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- Expected: ERROR 1701 - TRUNCATE cannot fire referential actions, so
(PES1UG23CS024@root@localhost) [transit_gap] > -- MySQL refuses it on any table a foreign key references.
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > TRUNCATE TABLE AGENCY;
ERROR 1701 (42000): Cannot truncate a table referenced in a foreign key constraint ('transit_gap`.`route`, CONSTRAINT `f
k_route_agency`)
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- DELETE on the same table is allowed, because it can cascade.
(PES1UG23CS024@root@localhost) [transit_gap] > -- (Do not run this unless you want the whole schema emptied.)
(PES1UG23CS024@root@localhost) [transit_gap] > -- DELETE FROM AGENCY;
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] >
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
```

Database Management Systems

Verify the final state - agencies, wards, stops and routes

```
SELECT * FROM AGENCY;
SELECT * FROM WARD;
SELECT * FROM STOP;
SELECT * FROM ROUTE;
SELECT * FROM COMMUTER;
SELECT * FROM JOURNEY;
SELECT * FROM FARE_RECEIPT;

SELECT * FROM COMMUTER_AGE;
SELECT * FROM WARD_DENSITY;
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > -- VERIFY THE FINAL STATE
(PES1UG23CS024@root@localhost) [transit_gap] > -- =====
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM AGENCY;
(PES1UG23CS024@root@localhost) [transit_gap] >
+-----+-----+-----+-----+-----+
| Agency_ID | Agency_Name | Mode | Headquarters_Location | Fleet_Size |
+-----+-----+-----+-----+-----+
| BMRL | Bangalore Metro Rail Corporation Ltd | METRO | Swami Vivekananda Road | 350 |
| BMTCL | Bengaluru Metropolitan Transport Corporation | BUS | Shanthinagar | 6000 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM WARD;
+-----+-----+-----+-----+-----+
| Ward_ID | Ward_Name | Zone | Population | Area_SqKm |
+-----+-----+-----+-----+-----+
| W150 | Jayanagar | UNASSIGNED | 42000 | 3.400 |
| W151 | Bellandur | UNASSIGNED | 68000 | 9.100 |
| W152 | Hebbal | UNASSIGNED | 51000 | 5.600 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM STOP;
+-----+-----+-----+-----+-----+-----+
| Agency_ID | Stop_Code | Stop_Name | Latitude | Longitude | Ward_ID |
+-----+-----+-----+-----+-----+-----+
| BMRL | S001 | Jayanagar Metro | 12.926000 | 77.580000 | W150 |
| BMTCL | S001 | Jayanagar 4th Block | 12.925100 | 77.583400 | W150 |
| BMTCL | S002 | Bellandur Gate | 12.926500 | 77.677000 | W151 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli x + v
(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM STOP;
+-----+-----+-----+-----+-----+-----+
| Agency_ID | Stop_Code | Stop_Name | Latitude | Longitude | Ward_ID |
+-----+-----+-----+-----+-----+-----+
| BMRL | S001 | Jayanagar Metro | 12.926000 | 77.580000 | W150 |
| BMTCL | S001 | Jayanagar 4th Block | 12.925100 | 77.583400 | W150 |
| BMTCL | S002 | Bellandur Gate | 12.926500 | 77.677000 | W151 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM PLATFORM;
+-----+-----+-----+-----+-----+
| Agency_ID | Stop_Code | Platform_Number | Wheelchair_Access |
+-----+-----+-----+-----+-----+
| BMRL | S001 | 1 | Y |
| BMRL | S001 | 2 | Y |
| BMTCL | S001 | 1 | Y |
| BMTCL | S001 | 2 | N |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

(PES1UG23CS024@root@localhost) [transit_gap] > SELECT * FROM ROUTE;
+-----+-----+-----+-----+-----+-----+-----+
| Route_ID | Route_Number | Route_Name | Route_Type | Total_Distance_KM | Agency_ID | Trunk_Route_ID |
+-----+-----+-----+-----+-----+-----+-----+
| R501 | S01A | Jayanagar Feeder | FEEDER | 6.20 | BMTCL | NULL |
| R502 | S02B | Bellandur Feeder | FEEDER | 7.80 | BMTCL | NULL |
| R001 | PURPLE | Purple Line | METRO | 43.00 | BMRL | NULL |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Write the server's response down first, then run it

1. `INSERT INTO STOP VALUES ('KSRTC','S999','Nowhere',12.9,77.5,'W150');`
Which error appears, and which constraint raises it?
2. `UPDATE AGENCY SET Agency_ID = 'BMTC-N' WHERE Agency_ID = 'BMTC';`
What happens to ROUTE and STOP, and what makes it happen?
3. `DELETE FROM AGENCY WHERE Agency_ID = 'BMRCL';` Name every table that loses rows.
4. `UPDATE WARD SET Zone = 'UNASSIGNED';` with no WHERE clause - what does the server report back?
5. Why does MySQL reject a subquery over PASS_TYPE inside an UPDATE on PASS_TYPE, and how is it rewritten so that it runs?
6. `TRUNCATE TABLE AGENCY;` is refused where `DELETE FROM AGENCY` is not. Why?

Database Management Systems

Quick check - choose one



1. Which statement is DML and reports the number of rows affected?
A) TRUNCATE B) DROP C) DELETE D) ALTER
2. 'Rows matched: 3 Changed: 3' is reported by
A) INSERT B) UPDATE C) SELECT D) DELETE
3. A second FARE_RECEIPT for journey J001 is rejected by
A) a foreign key B) a CHECK C) a UNIQUE constraint D) NOT NULL
4. Deleting trunk route R500 leaves R501 and R502 in place because fk_route_trunk uses
A) ON DELETE CASCADE B) ON DELETE SET NULL C) ON DELETE NO ACTION D) ON UPDATE CASCADE

Quick check

1 - C 2 - B 3 - C 4 - B

Predict the outcome

1. ERROR 1452 from fk_stop_agency - no agency KSRTC exists, so the child row is rejected.
2. fk_route_agency and fk_stop_agency are ON UPDATE CASCADE, so ROUTE and STOP are rewritten to BMTC-N without being touched.
3. STOP loses the BMRCL stops, PLATFORM loses their platforms, ROUTE loses RM01, and SERVES loses any pairing that used them.
4. Query OK, 3 rows affected - Rows matched: 3, Changed: 3. Matched counts what WHERE selected; changed counts what actually differed.
5. ERROR 1093 - the target table may not appear in the FROM clause. Wrap the subquery in a derived table and the same statement runs.
6. TRUNCATE does not process rows one at a time, so it cannot fire the referential actions.



THANK YOU

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